(c) Alcohol (d) Paraffin

Chhattisgarh P.C.S. (Pre) 2005

Ans. (c)

Liquor or alcoholic beverages are prepared by fermentation of different substances like molasses, etc. The percentage of alcohol also varies in them. Beer, Champagne, Cider, Port and Sherry, Brandy, Whisky, Rum, Gin etc. are some types of liquors.

S. No.	Name	Alcohol % ≈	Raw Material
1.	Rum	45 to 55%	Molasses
2.	Brandy	40 to 50%	Grapes
3.	Whisky	40 to 50%	Barley, Corn
4.	Beer	3 to 6%	Barley
5.	Champagne	10 to 15%	Grapes
6.	Cider	2 to 6%	Apple

7. Alcohol that is derived from fermentation of germinated barley grains is known as:

(a) Beer (b) Wine

(c) Vodka (d) Rum

U.P. R.O./A.R.O. (Mains) 2016

Ans. (a)

Beer is usually derived from the fermentation of malt derived from the digestion of germinated barley grains. Barley, water, hops and yeast are the four magic ingredients that are required for making beer. Other grains like maize, rice, rye and wheat are also used in making beer. Whisky is also made from fermented barley grains.

8. The breath test conducted by police to check drunken driver has which one of the following on the filter paper?

(a) Potassium dichromate-sulfuric acid

(b) Potassium permanganate-sulfuric acid

(c) Silica gel coated with silver nitrate

(d) Turmeric

(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (a)

The breath test conducted by police to check drunken driver through old breathalyzer has potassium dichromate-sulfuric acid on the filter paper. When alcohol vapour makes contact with the orange dichromate coated crystals, the colour changes from orange to green due to oxidation of alcohol into acetic acid. The degree of the colour changes is directly related to alcohol level in the breath.

C. Polymer

Notes

Polymerization :

- In polymer chemistry, polymerization is a process of reacting monomer molecules together in a chemical reaction to form polymer chains or three dimensional networks.

Natural polymers :

1. **Sporopollenin :** Sporopollenin is one of the most chemically inert biological polymers. It is a major component of the tough outer (exine) walls of plant spores and pollen grains. It is chemically very stable and is usually well preserved in soils and sediments. The chemical composition of sporopollenin is not exactly known, due to its unusual chemical stability and resistance to degradation by enzymes and strong chemical reagents. Analyses have revealed a mixture of biopolymers, containing mainly long chain fatty acids, phenylpropanoids, phenolics and traces of carotenoids.
2. **Protein :** Protein is the polymer of amino acids. They are large biomolecules or macromolecules, consisting of one or more long chain of amino acid residues. They are building material of organism's body and silk, wool, hair etc.
3. **Starch, cellulose, glycogen** are the polymers of glucose.
4. **Chitin :** Chitin is the polymer of N-acetyl amino glucose which contains nitrogen. It is hard, tough substance that occurs widely in nature, particularly in the exoskeleton of arthropods such as crabs, insects and spiders. The walls of hyphae (microscopic filaments of fungi) are composed of chitin.
5. **Nucleic acid (DNA/RNA) :** Polymers of nucleotides.
6. **Natural Rubber :** Natural rubber is the polymer of organic compound isoprene. Natural rubber is harvested mainly in the form of latex from the rubber tree and some other trees such as – Dandelion, Spurges, gutta percha etc.
7. Wood, cotton, silk, wool, leather, enzymes and cellulose are all examples of polymers.

Artificial polymers :

- Some polymers of unsaturated hydrocarbons are as follows :

(a) Polyethylene or Polythene :

- Polythene is the polymer of **ethylene**. It is the most common plastic.
- Polythene is thermoplastic, non-conductor of electricity. It has a high ductility and impact strength as well as low friction.
- Polythene is water resistant.
- Polythene is used in packaging, plastic bags, plastic films, geomembranes, containers including bottles etc.

(b) Teflon :

- Teflon is also known as polytetrafluoroethylene **(PTFE)**.
- Teflon is a synthetic fluoropolymer of tetrafluoroethylene $(C_2F_4)_n$.
- Chemically it is unreactive and heat resistance.
- Melting point of teflon is 327°C.
- Teflon is used as a non-stick coating for pans and other cookware because it is very smooth.
- Teflon is non-reactive, partly because of the strength of carbon-fluorine bonds, and so it is often used in containers and pipe work for reactive and corrosive chemicals such as conc. nitric acid, conc. sulphuric acid, conc. hydrochloric acid, aqua regia and strong sodium hydroxide.

(c) Polyvinyl chloride (PVC) :

- Polyvinyl chloride is a synthetic thermoplastic polymer made by polymerizing vinyl chloride.
- Polyvinyl chloride is non-conductor of electricity, ductile and waterproof.
- Flexible forms of PVC are used in preparation of pipes, insulation, shoes, garments etc., while rigid PVC is used for moulded articles.
- PVC contains dangerous chemical additives including phthalates, lead, cadmium and/or organotins, which can be toxic to child's health. These toxic additives can leach out or evaporate into the air overtime, posing unnecessary dangers to children.

(d) Neoprene :

- Neoprene is also called polychloroprene or pc-rubber. It is a family of synthetic rubbers that are produced by polymerization of chloroprene.
- Neoprene exhibits good chemical stability and maintains flexibility over a wide temperature range.
- Neoprene is used to prepare pipes, belts and other things.

(e) Polypropylene :

- It is produced through chain-growth polymerization of the monomer propylene.
- It is also known as polypropene. It is a thermoplastic polymer used in a wide variety of applications.
- Its properties are similar to polyethylene, but it is still harder and more heat resistant.
- Polypropylene is the second-most widely produced plastic (after polyethylene) and it is often used in packaging and labeling.
- It belongs to the group of polyolefins and is partially crystalline and non-polar with a high chemical resistance.

Question Bank

1. Natural rubber is a polymer of –

(a) Butadiene (b) Ethylene
(c) Isoprene (d) Styrene

Jharkhand P.C.S. (Pre) 2013
U.P.P.C.S. (Pre) 1992

Ans. (c)

Polymerization is the process of joining together a large number of small molecules to make a very large molecule. The reactants (i.e. the small molecules from which the polymer is constructed) are called monomers and products of the polymerization process are called polymers. Natural rubber is the natural polymer of isoprene. Isoprene is a colourless liquid made by destructive distillation of petroleum.

2. Natural rubber is a polymer of :

(a) Isoprene (b) Styrene
(c) Vinyl acetate (d) Propene
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020
65th B.P.S.C. (Pre) 2019

Ans. (a)

See the explanation of above question.

3. Which of the following is not a Natural polymer?

(a) Wool (b) Silk
(c) Leather (d) Nylon

U.P.P.C.S. (Mains) 2015
U.P.U.D.A./L.D.A. (Pre) 2001

Ans. (d)

Wool, silk, leather are the natural polymers but nylon is not a natural polymer.

4. Which one of the following is a natural polymer?

(a) Bakelite (b) Silk
(c) Kevlar (d) Lexan

U.P.P.C.S. (Pre) 2017

Ans. (b)

See the explanation of above question.

5. Which of the following is not a natural polymer –

(a) Ghee (b) Starch
(c) Protein (d) Cotton

U.P.Lower Sub. (Pre) 2009

Ans. (a)

Ghee is not a polymer while rest all are the natural polymers.

6. **Which among the following is a constituent of natural silk?**

(a) Potassium (b) Magnesium
(c) Nitrogen (d) Phosphorus

U.P. P.C.S. (Pre) 2023

Ans. (c)

Silk is a naturally occurring protein fiber produced from the larvae of the moth to build their cocoons. This fiber is used for the commercial production of silk fabric. Nitrogen is a constituent element of natural silk.

7. **Cellulose and starch both are made up of :**

(a) (+) – glucose (b) (–) – fructose
(c) Both (a) and (b) given above (d) (+) – galactose

Uttarakhand P.C.S. (Pre) 2016

Ans. (a)

Cellulose is the most commonly found organic compound, biopolymer and polysaccharide on the earth. Each of these molecules contains several hundred to thousands glucose molecules in the form of an inosculated homopolymer series. Starch plants have structural polysaccharides which contain two types of homopoly-saccharide molecules made of glucose cells – about 10 to 30 percent amylase and about 70 to 90 percent amylopectin molecules.

8. **The most abundantly found organic compound in nature is –**

(a) Glucose (b) Fructose
(c) Sucrose (d) Cellulose

U.P.P.C.S. (Mains) 2014
U.P.P.C.S. (Mains) 2012

Ans. (d)

Cellulose is a form of carbohydrate. It is a polysaccharide consisting of a linear chain of several hundred to many thousand of β linked D-glucose units. It is the chief constituent of cell walls in living organisms. Wood is mostly cellulose, making cellulose the most abundant type of organic compound on the earth. Its purest natural form is cotton.

9. **To which of the following chemical groups – Cellulose belongs?**

(a) Carbohydrates (b) Proteins
(c) Lipids (d) Nucleic acids

Uttarakhand P.C.S. (Pre) 2021

Ans. (a)

See the explanation of above question.

10. **Which of the following is an example of a non-cellulosic fibre?**

(a) Rayon (b) Linen
(c) Jute (d) Nylon

R.A.S./R.T.S. (Pre) 2018

Ans. (d)

Rayon, cotton, hemp, jute and linen are made of cellulose, while nylon, polyesters etc. are non-cellulosic fibres.

11. **Select the incorrect statement out of the following :**

(a) Cotton is suitable for use as clothing in summer because it absorbs moisture.
(b) Polycarbonate is used for making CDs.
(c) Acrylic is also called artificial silk as it is prepared from cotton but has shine like silk.
(d) Teflon is used for coating non-stick kitchenwares.
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2022

Ans. (c)

Rayon is soft, lustrous and absorbent which exactly copies the characteristics of silk. Rayon appears and feels like silk. Hence, Rayon is called artificial silk. Fibre obtained by chemically treating wood pulp is called rayon or artificial silk. Acrylic fibers are the synthetic fibers which resemble natural wool. Statements of other options are correct.

12. **Which of the following polymer is NOT a thermoplastic?**

(a) Teflon (b) Neoprene
(c) Polystyrene (d) Polythene

U.P. P.C.S. (Pre) 2020

Ans. (b)

Among the given options, neoprene is not a thermoplastic. Neoprene is a family of synthetic rubbers that are produced by polymerization of chloroprene. It is used to prepare pipes, belts and other things. It is also called polychloroprene or pc-rubber.

13. **Which one of the following polymers is not biodegradable?**

(a) Cellulose (b) Starch
(c) Protein (d) PVC

U.P. R.O./A.R.O. (Pre) 2017

Ans. (d)

PVC (Polyvinyl chloride) is a synthetic polymer of vinyl chloride which is not biodegradable while cellulose, starch and protein are natural polymers and are biodegradable.

14. Which one of the following is used in the synthesis of polythene?

(a) Methane (b) Ethane

(c) Propane (d) Butane

R.A.S./R.T.S.(Pre) 2008

Ans. (*)

Polythene is produced by the polymerization of ethylene. Hence it is a polymer of ethelyne.

15. Which one of the following gases is used in manufacturing of polythene?

(a) Ethylene (b) Carbon-di-oxide

(c) Nitrogen (d) Carbon monoxide

U.P. R.O./A.R.O. (Mains) 2021

Ans. (a)

See the explanation of above question.

16. Which gas is obtained from plastic ?

(a) Polynitrogen (b) Polyhydrogen

(c) Polychlorine (d) Polyethylene

Uttarakhand P.C.S. (Pre) 2010

Ans. (d)

Polyethylene gas is produced from plastic.

17. Which one of the following is an example of melamine used for making floor tiles ?

(a) Polyethene

(b) Thermoplastics

(c) PVC

(d) Thermosetting plastic

70th B.P.S.C. (Pre) 2024

Ans. (d)

Melamine, a versatile thermosetting plastic, is used to make a wide variety of products, including kitchenware, floor tiles, laminates, adhesives, and fire-retardant materials, as well as in the production of melamine resins and foams. Melamine is also used in the production of particle boards, which are used in furniture, flooring, and other construction applications.
Thermosetting plastics, also known as thermosets, are polymers that become permanently rigid and fixed when heated and undergo an irreversible chemical change, meaning they cannot be reshaped or remelted once cured.

18. With reference to polyethylene terephthalate, the use of which is so widespread in our daily lives, consider the following statements :

1. Its fibres can be blended with wool and cotton fibres to reinforce their properties.

2. Containers made of it can be used to store any alcoholic beverage.

3. Bottles made of it can be recycled into other products.

4. Articles made of it can be easily disposed of by incineration without causing greenhouse gas emissions.

Which of the statements given above are correct?

(a) 1 and 3 (b) 2 and 4

(c) 1 and 4 (d) 2 and 3

I.A.S. (Pre) 2022

Ans. (a)

Polyethylene terephthalate (PET or PETE), is the most common thermoplastic polymer resin of the polyester family and is used in fibres for clothing, containers for liquids and foods, and thermoforming for manufacturing, and in combination with glass fibre for engineering resins. It is often used in durable - press blends with other fibres such as rayon, wool, and cotton, reinforcing the inherent properties of those fibres.
PET bottles are not considered safe to be used to store alcoholic beverages. Maharashtra government has banned the sale of alcohol in PET bottles saying that plastic packaging is dangerous to human health and the country liquor and country made foreign liquor cannot be sold in such bottles.
PET bottles can be recycled into other products. Some everyday items recycled plastic bottles can be made into are : Plastic packaging, Clothing and shoes, Carpets and soft furnishings, Furniture, Automotive parts etc.
PET, like many plastics, is also an excellent candidate for thermal disposal (incineration), as it is composed of carbon, hydrogen, and oxygen, with only trace amounts of catalyst elements (but no sulfur). However, incineration of PET causes greenhouse gas emissions.

19. In which of the following are hydrogels used ?

1. Controlled drug delivery in patients

2. Mobile air-conditioning systems

3. Preparation of industrial lubricants

Select the correct answer using the code given below:

(a) 1 only (b) 1 and 2 only

(c) 2 and 3 only (d) 1, 2 and 3

I.A.S. (Pre) 2024

Ans. (d)

Hydrogels are polymeric materials consisting of a sparse network of polymer chains embedded in an aqueous medium. Hydrogels can retain large amounts of water within their intermolecular space due to strong hydrophilicity of the polymer chains and large porosity. Flexibility of hydrogels, which is because of their water content, makes it possible to use them in different conditions ranging from industrial to biological.

Hydrogels considerably enhance the therapeutic outcome of drug delivery and have found enormous clinical use. Their porosity and compatibility with aqueous environments make them highly attractive bio-compatible drug delivery vehicles. Owing to their tunable physical properties, controllable degradability and capability to protect labile drugs from degradation, hydrogels serve as a platform on which various physiochemical interactions with the encapsulated drugs occur to control drug release. Hence, point 1 is correct.
Hydrogels are flexible and soft 3-dimensional networks with high water content and evaporative and radiative cooling properties that make them suitable for use in passive cooling technology. Hydrogels can be used in mobile air conditioning systems as desiccants to control humidity. Hence, point 2 is correct.
Hydrogels can be used in industrial lubricants to improve their properties, such as reducing friction and wear, controlling viscosity, and improving thermal stability. Hydrogels have played a crucial role as water-based lubricants in recent years. Hydrogel lubricants not only have the advantage of eliminating lubrication failure compared to lubricating liquids due to their non- flowability but also show potential applications in the field of underwater lubrication and bio-lubrication. Hence, point 3 is also correct.

20. Which one of the following polymer is used in making bullet-proof vests ?
(a) Bakelite (b) Polyamides
(c) Teflon (d) Polyurethanes
U.P.P.C.S. (Mains) 2014
Ans. (b)

Kevlar is a material commonly used to make lightweight bulletproof helmets and vests. Kevlar is a para-aromatic polyamide (Poly-Paraphenylene Terephthalamide) synthetic fibre. It contains lots of inter-chain bonds which makes it extremely strong. Layers of laminated glass are also used for making bulletproof materials.

21. Light weight bullet proof helmets and jackets use which of the following fibres for their production?
(a) Terylene (b) Nylon
(c) Kevlar (d) None of the above
70th B.P.S.C. (Pre) (Re-Exam) 2024
Ans. (c)

See the explanation of above question.

22. What is the 'fibre' used to make bulletproof jackets?
(a) Nylon (b) Terylene
(c) Tweed (d) Kevlar
69th B.P.S.C. (Pre) 2023
Ans. (d)

See the explanation of above question.

23. Which of the following polymer is used in the manufacture of bulletproof material?
(a) Nylon 6, 6 (b) Rayon
(c) Kevlar (d) Dacron
Jharkhand P.C.S. (Pre) 2010
Ans. (c)

See the explanation of above question.

24. Which one of the following polymers is widely used for making bulletproof material ?
(a) Polyvinyl chloride (b) Polyamides
(c) Polyethylene (d) Polycarbonates
U.P. U.D.A./L.D.A. (Mains) 2010
U.P.P.S.C. (GIC) 2010
U.P.P.C.S. (Spl.) (Pre) 2005
I.A.S. (Pre) 1995
Ans. (b)

See the explanation of above question.

25. Which one of the following polymers is used for making bullet-proof windows?
(a) Polycarbonates (b) Polyurethanes
(c) Polystyrene (d) Polyamides
U.P.P.C.S. (Pre) 2015
Ans. (a)

Bullet-resistant glass is made or manufactured by using polycarbonate, thermoplastic and layers of laminated glass. It can be used in making bulletproof jackets.

26. Which one of the following is used in making 'Bullet-proof Jacket'?
(a) Fibrous glass (b) Gun metal
(c) Lead (d) Laminated glass
Uttarakhand P.C.S. (Pre) 2005
Ans. (d)

See the explanation of above question.

27. Which one of the following polymers are used for making bulletproof materials?
I. Kevlar **II. Glyptal**
III. Lexan
Select the correct answer using the codes given below :
Code :
(a) I and II (b) II and III
(c) I and III (d) None of the above
U.P.P.C.S. (Mains) 2017
Ans. (c)

Kevlar (polyamide - best known for its use in ballistic and stab-resistant body armour) and Lexan (polycarbonate) are used for making bulletproof materials. Glyptal is a condensation polymer which is used in the manufacturing of paints and lacquers (protective coating for woods, metals).

28. Consider the following statements :

1. Teflon and Dacron are polymers

2. Neoprene is synthetic rubber

3. Polythene is polyethylene polymer

4. Natural rubber is chloroprene

Which of the above statements are correct :

(a) 1, 2 and 3 (b) 1, 2 and 4

(c) 2, 3 and 4 (d) 1, 3 and 4.

U.P.P.C.S. (Pre) 2003

U.P. U.D.A./L.D.A. (Pre) 2002

Ans. (a)

Natural rubber is a polymer of 'Isoprene' which is obtained as a latex from rubber trees. Rubber obtained from an artificial source is known as synthetic rubber. Neoprene is a family of synthetic rubbers that are produced by polymerization of chloroprene. Teflon and Dacron are examples of polymers. Polythene or polyethylene is a polymer of ethylene.

29. A polymer used for making nonstick surface coating for utensils is –

(a) Polyvinyl chloride (b) Teflon

(c) Polystyrene (d) Polypropylene

U.P.P.C.S. (Spl.) (Mains) 2008

Ans. (b)

Polytetrafluoroethylene (PTFE) is usually known as Teflon. PTFE is a solid fluorocarbon. Its density is 2.2 g/cm^3 and its melting point is 327^{o}C. This is especially used for making a non-stick surface coating for utensils.

30. Which one of the following polymers is used for the production of non-stick pans?

(a) Teflon (b) Neoprene

(c) P.V.C. (d) Gutta-Percha

U.P.P.C.S. (Pre) 2015

Ans (a)

See the explanation of above question.

31. Non-stick frying pans are coated with :

(a) Orlon (b) Teflon

(c) Polystyrene (d) Polypropylene

Jharkhand P.C.S. (Pre) 2016

Ans. (b)

See the explanation of above question.

32. What is Teflon?

(a) Fluorocarbon (b) Hydrocarbon

(c) Microbicides (d) Insecticide

Jharkhand P.C.S. (Pre) 2010

Ans. (a)

See the explanation of above question.

33. Teflon is a polymer of which of the following monomers?

(a) Tetrafluoroethylene

(b) Vinyl chloride

(c) Chloroprene

(d) Acetylene dichloride

(e) None of the above/ More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (a)

Polytetrafluoroethylene commonly known as Teflon is a polymer of tetrafluoroethylene.

34. Teflon is the common name of –

(a) Polytetrafluoroethylene

(b) Polyvinyl chloride

(c) Polyvinyl fluoride

(d) Dichlorodifluoro methane

Uttarakhand P.C.S. (Pre) 2016

Ans. (a)

See the explanation of above question.

35. Which among the following is not a polymer ?

(a) Nylon (b) Teflon

(c) Caprolactam (d) Polystyrene

Uttarakhand P.C.S. (Pre) 2012

Ans. (c)

The polymer is a long chain molecule made up of many small identical units. Polymers are common in nature. Wood, rubber, cotton, silk, proteins, enzymes and cellulose are all examples of polymers. A wide variety of synthetic polymers has been produced largely from petroleum-based raw materials. These include polyurethane, teflon, polyethylene, polystyrene, and nylon. Caprolactam (CPL) is an organic compound with the formula $(CH_2)_5C(O)NH$. It is being used as a raw material for nylon.

36. Which one of the following substance is NOT synthetic?

(a) Fibroin (b) Lexan

(c) Neoprene (d) Teflon

U.P.P.C.S. (Pre) 2020

Ans. (a)

Among the given options fibroin is a natural substance and not synthetic. Fibroin is an insoluble protein present in silk produced by numerous insects, such as the larvae of Bombyx mori, and other moth genera. Silk in its raw state consists of two main proteins, sericin and fibroin.

37. With reference to perfluoroalkyl and polyfluoroalkyl substances (PFAS) that are used in making many consumer products, consider the following statements:

1. PFAS are found to be widespread in drinking water, food and food packaging materials.

2. PFAS are not easily degraded in the environment.

3. Persistent exposure to PFAS can lead to bioaccumulation in animal bodies.

Which of the statements given above are correct?

(a) 1 and 2 only (b) 2 and 3 only

(c) 1 and 3 only (d) 1, 2 and 3

I.A.S. (Pre) 2024

Ans. (d)

Perfluoroalkyl and polyfluoroalkyl substances (PFAS) are a large, complex group of synthetic chemicals that have been widely used in consumer products around the world. Some of the major industry sectors using PFAS include aerospace and defence, automotive, food contact materials, cosmetics, textiles, leather and apparel, construction and household products, electronics, firefighting, food processing, and medical articles etc. PFAS are found widespread in drinking water, foods, food packaging materials and other consumer products. PFAS molecules have a chain of linked carbon and fluorine atoms. Because the carbon-fluorine bond is one of the strongest, these chemicals do not degrade easily in the environment. Hence, statements 1 and 2 are correct.

Over time, PFAS may leak into the soil, water, and air. People are most likely exposed to these chemicals by consuming PFAS-contaminated water or food, using products made with PFAS, or breathing air containing PFAS. Over time, people may take in more of these chemicals than they excrete, a process that leads to bioaccumulation in bodies. Because of their widespread use and their persistence in the environment, many PFAS are found in the blood of people and animals all over the world. Hence, statement 3 is also correct.

38. Bisphenol A (BPA), a cause of concern, is a structural/ key component in the manufacture of which of the following kinds of plastics?

(a) Low-density polyethylene (b) Polycarbonate

(c) Polyethylene terephthalate (d) Polyvinyl chloride

I.A.S. (Pre) 2021

Ans. (b)

Bisphenol A (BPA) is a chemical produced in large quantities for use primarily in the production of polycarbonate plastics and epoxy rasins. It is found in various products including shatterproof windows, food packaging materials, eyewear, water bottles, and epoxy resins that coat some metal food cans, bottle tops, and water supply pipes. BPA has been used in food packaging since the 1960s. The primary source of exposure to BPA for most people is through the diet. While air, dust, and water are other possible sources of exposure. BPA in food and beverages accounts for the majority of daily human exposure. Bisphenol A can leach into food from the protective internal epoxy resin coatings of canned foods and from consumer products such as polycarbonate tableware, food storage containers, water bottles, and baby bottles. Exposure to Bisphenol A may cause or contribute to a variety of health problems including brain disorder, infertility, heart disease, type 2 diabetes etc.

39. What is Bisphenol A (BPA)?

(a) A medical test for detecting cancer

(b) A test for testing the use of drugs to improve performance by athletes

(c) A chemical used for the development of food-packaging materials

(d) A special type of alloy steel

I.A.S. (Pre) 2008

Ans. (c)

See the explanation of above question.

40. Phenol is used in the manufacture of which one of the following ?

(a) P.V.C. (b) Nylon

(c) Polystyrene (d) Bakelite

U.P.P.C.S. (Mains) 2010

Ans. (d)

Bakelite is a thermosetting phenol formaldehyde resin, formed from a condensation reaction of phenol with formaldehyde. It was developed by the Belgian-American chemist Leo Baekeland in 1907.

41. Bakelite is formed by the condensation of :

(a) Urea and formaldehyde

(b) Phenol and formaldehyde

(c) Phenol and acetaldehyde

(d) Melamine and formaldehyde

(e) None of the above/ More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (b)

See the explanation of above question.